#1-RADIOACTIVITY|NUCLEAR STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

Nuclear chemistry :

Radioactivity : Some nuclei of some elements are unstable and undergo nuclear decay or disintegration spontaneasly (without any external means or excitation).

Emission of nuclear particles like α-emission,



5:15 / 38:05

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nuclear decay or disintegration spontaneasly (withoutany external means or excitation).

Emission of nuclear particles like α-emission, β-emission and γ-rays emission from unstable nuclei is called radioactivity.



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5:56 / 38:05

Nuclear stability :

→ High stability of most nuclei of stable elements is due to the very strong nuclear forces that operate in within nucleus (or at nuclear distance) (of the order of 10^{-15} m or fm) { 1 fm = 10^{-15} m }



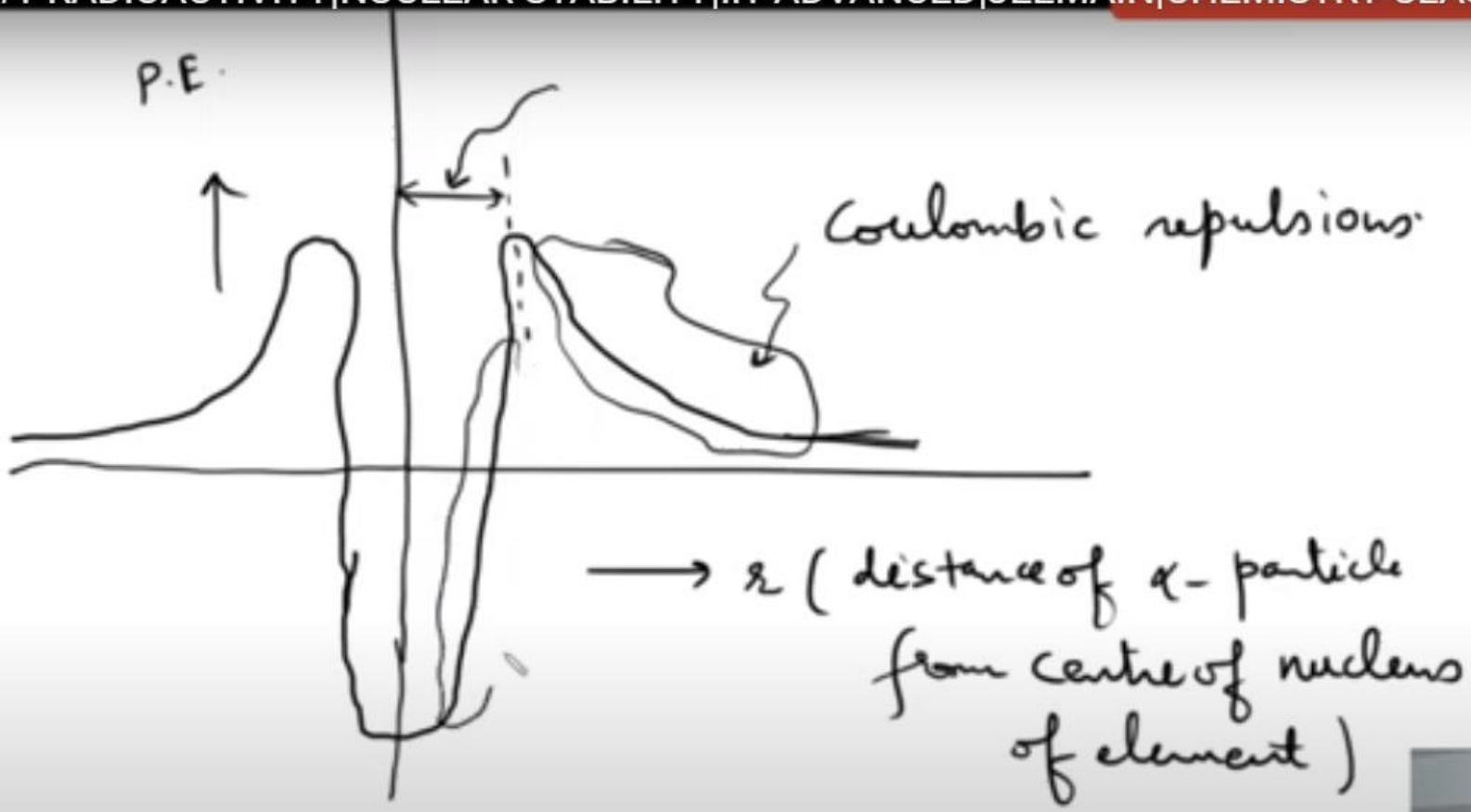
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→ Evidence of nuclear forces can be illustrated by α-particle scattering experiment or neutron scattering experiment.

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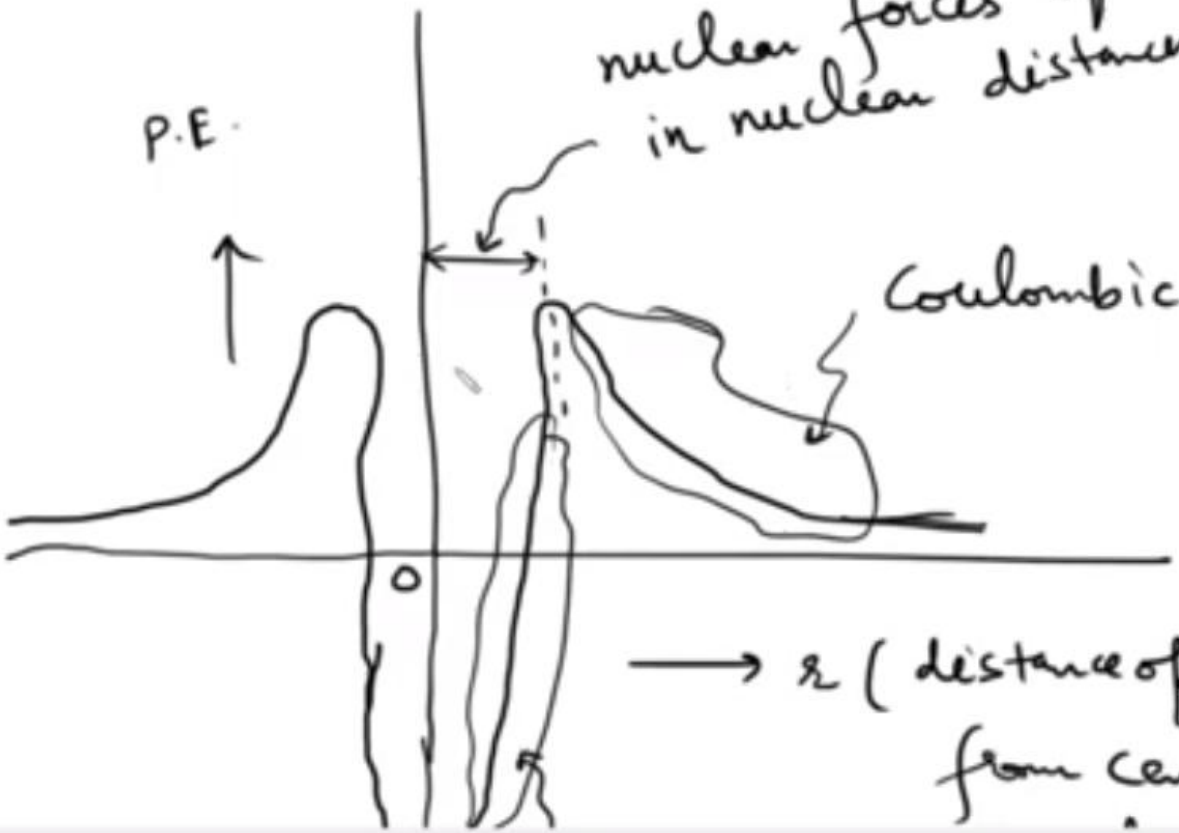
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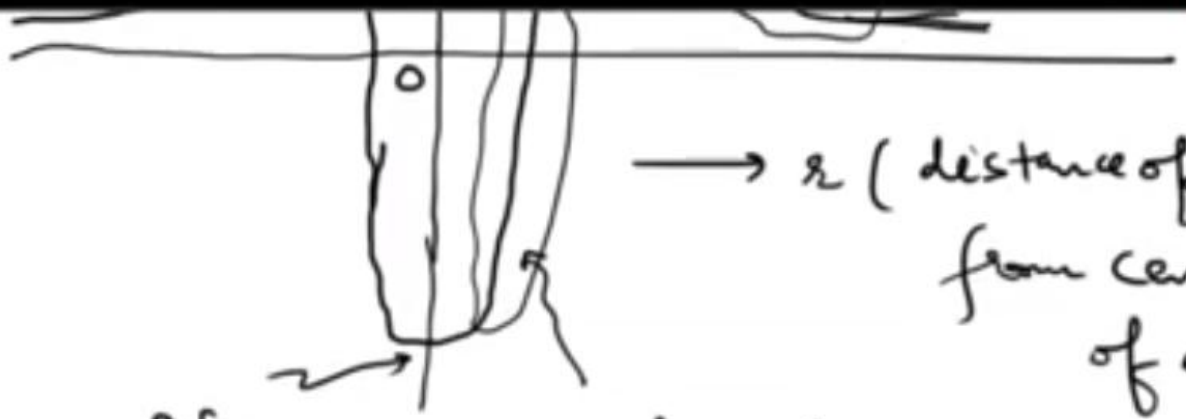
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13:12 / 38:05

experiment.



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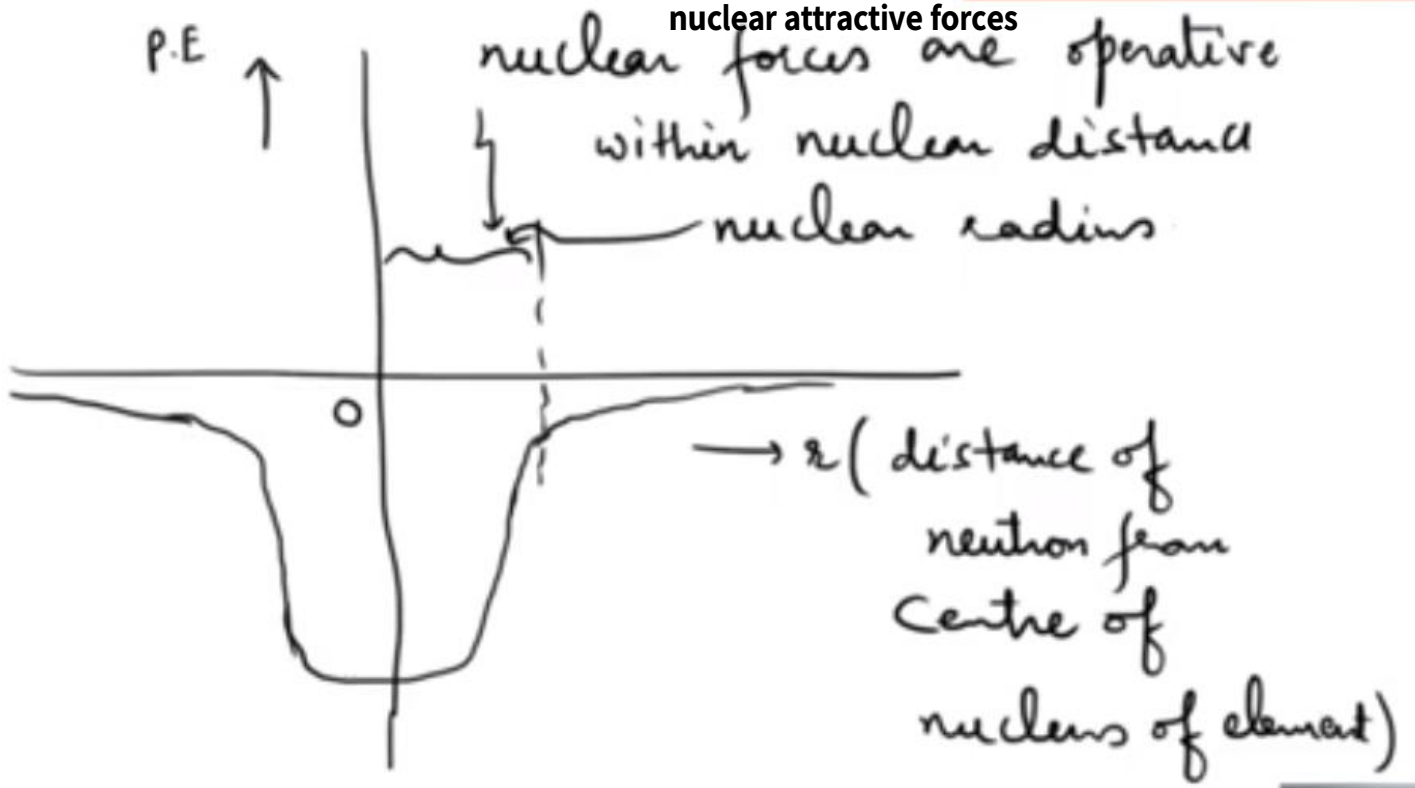


→ r (distance of α- particle from centre of nucleus of element)

-ve P.E (indicates attraction)

nuclear forces become dominant (attractive)

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P.E diagram of neutron scattering experi

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P.E diagram of neutron scattering experiment.

→ strongest nuclear force is nearly 137 times stronger than coulombic forces.

→ Infact nuclear force varies with distance and on an average it is 40-50 times stronger than coulombic forces.



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21:25 / 38:05

137 times stronger than coulombic forces.

→ Infact nuclear force varies with distance and on an average it is 40-50 times stronger than coulombic forces.

→ Yukawa proposed his meson-theory for stability of nucleus according to which there is rapid exchange of π-mesons between p-p, p-n, n-n and energy is released.



→ π - meson may be electrically neutral, +ve or -vely charged. (π^0, π^+, π^-) charge of [redacted] π^+, π^- meson is same as charge on electron but π - meson is much heavier than electron. nearly 270 times heavier



Mass defect and Binding energy :

→ It is experimentally found that atom of every element is slightly lighter in mass than the mass sum of its constituent particles.

e.g oxygen atom → $8p, 8n, 8e^-$



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Observation : mass of O^{16} atom

= 15.99 amu

(approx)

mass sum of $8p+8n+8e^-$ = 16.13 amu.

→ when protons and neutrons combine to form nucleus, there is a mass loss called mass defect and an equivalent



called mass defect and an equivalent amount of energy is released called called Binding energy.

$E =$



Binding energy:

$E = \Delta m \times c^2$

mass defect

Binding energy

V.Imp

1 amu ≅ 931.4 MeV

mass defect

Binding energy

(Remember)



defect Binding energy

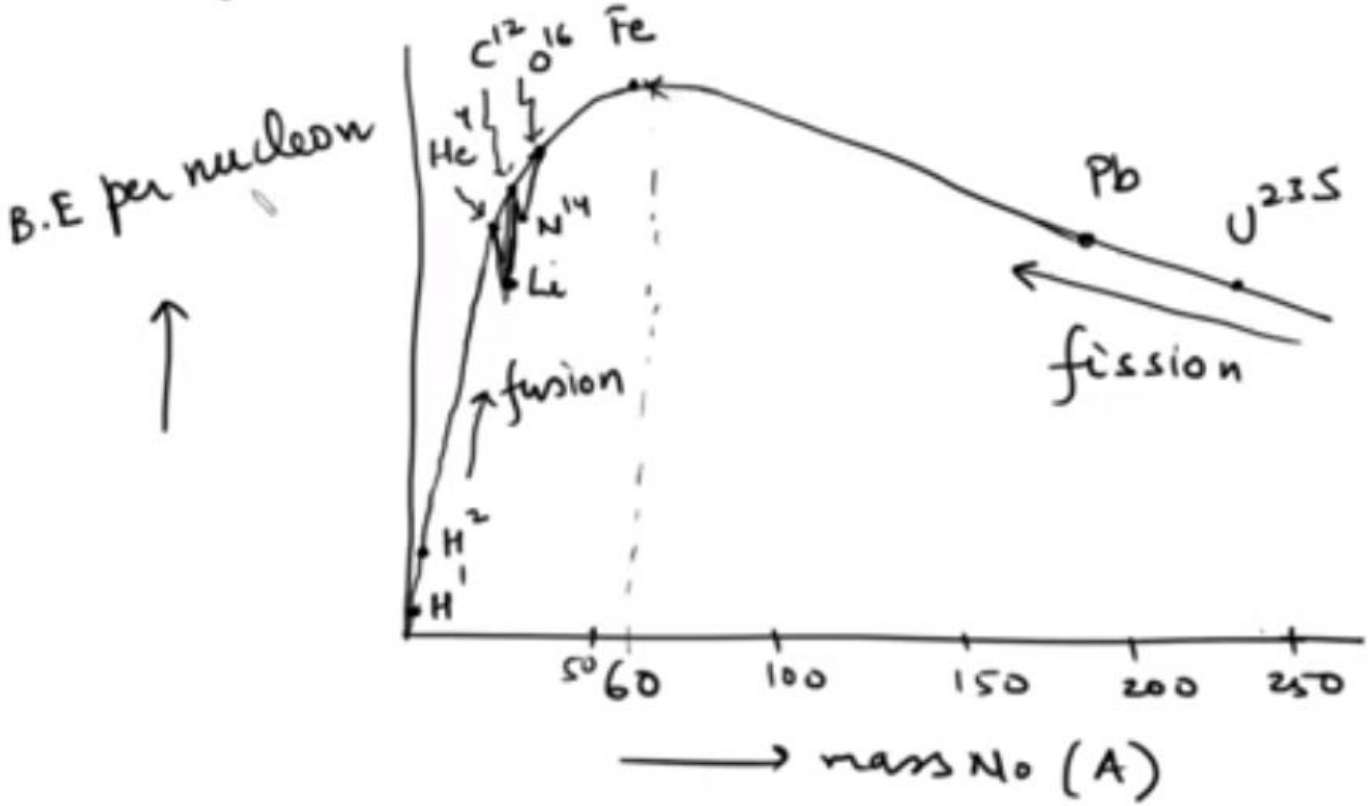
Binding energy.

(Remember)

Imp

Note: Higher is the binding energy per nucleon, greater will be the stability of nucleus.





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⟶ mass No (A)

→ B.E per nucleon is found to be maximum for mass number nearly 60.

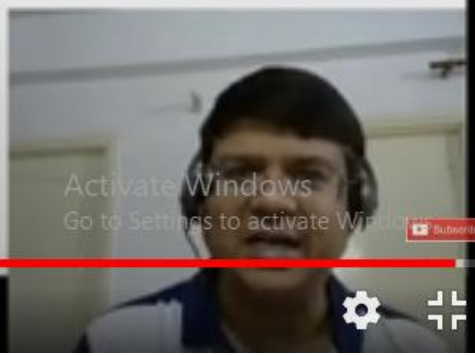
→ Lighter nuclei may undergo fusion while heavier nuclei may undergo

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#1-RADIOACTIVITY|NUCLEAR STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

mass number nearly 60.

→ Lighter nuclei may undergo fusion while heavier nuclei may undergo fission to get into stability zone under appropriate conditions.



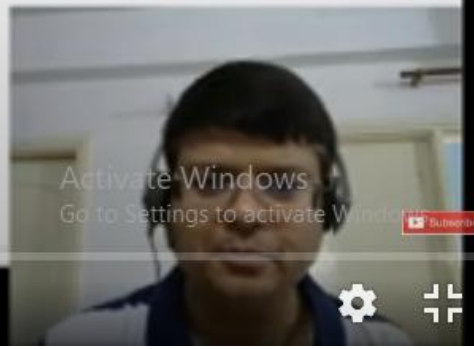
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#2-MAGIC NUMBER| BAND OF STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

Magic number :

→ It is observed that the isotopes where number of neutrons or no. of protons are even in number, they are more plentiful and stable nucleus than the odd no. of protons or neut



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2:41 / 30:00

in number, they are more plentiful and stable nucleus than the odd no. of protons or neutrons.

→ Particularly the isotopes where no. of protons or neutrons are equal to 2, 8, 20, 50, 82, 126, are found to

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protons or neutrons are equal to 2, 8, 20, 50, 82, 126, are found to be highly stable and hence these numbers are called magic Numbers

$_2He$, $_8O$, ~~$_{20}Ar$~~ $_{20}Ca$ etc are highly stable



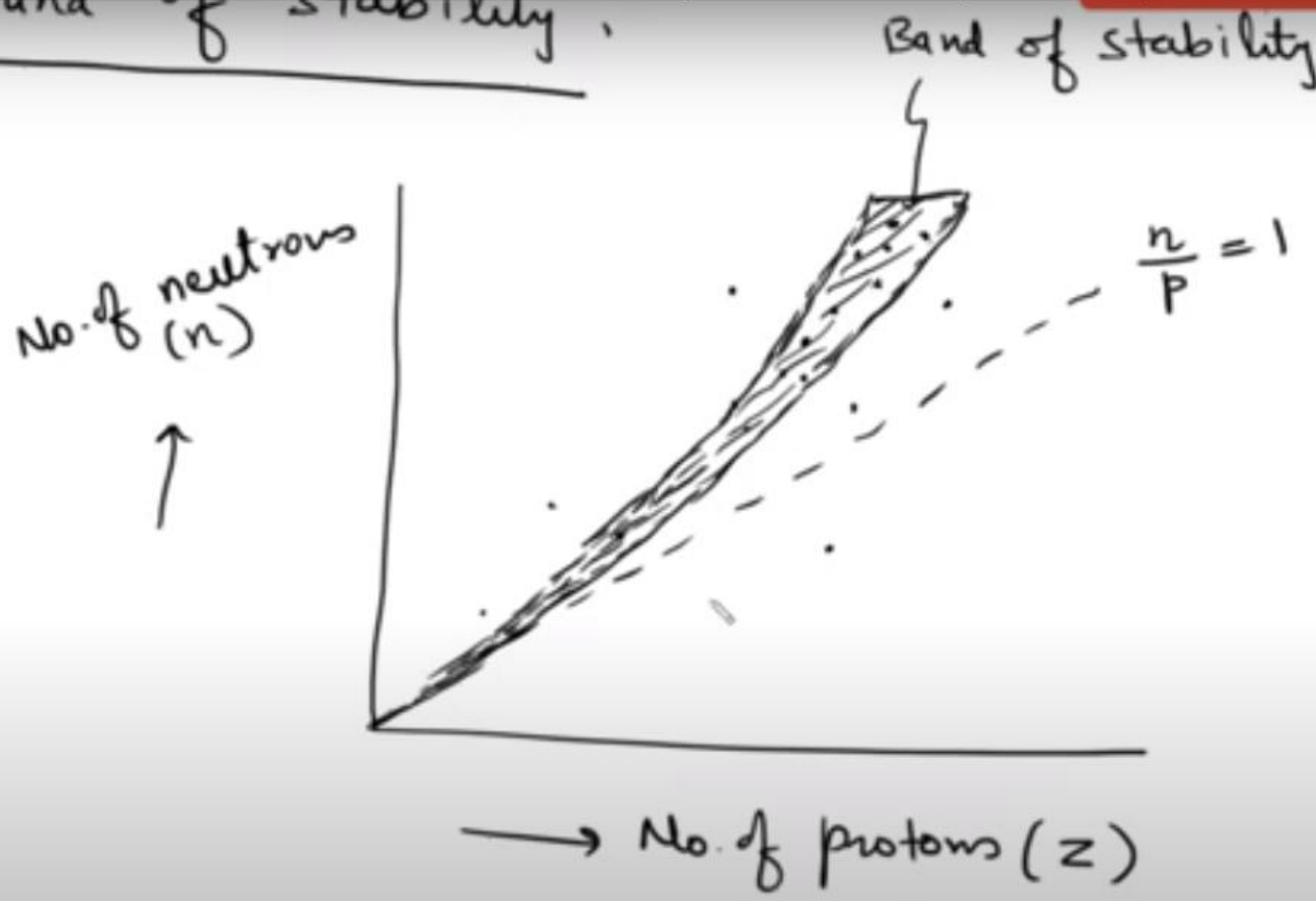
$_{2}He$, $_{8}O$, $_{20}Ca$ etc are highly stable.

(v.imp)

Band of stability :



#2-MAGIC NUMBER| BAND OF STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...



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→ No. of protons (Z)

→ Isotopes which lie in band of stability are found to be stable (non-radioactive)

→ For atomic No up $Z = 20$, $\frac{n}{p}$ for stable isotopes lie in range 1 to 1.1.

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#2-MAGIC NUMBER| BAND OF STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

→ For atomic No up $Z = 20$, $\frac{n}{p}$ for stable isotopes lie in range 1 to 1.1.

* For heavier isotopes, $\frac{n}{p}$ for stable isotopes may be as high as nearly 1.5.



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11:58 / 30:00

isotopes may be as high as nearly 1.5.

→ Unstable isotopes will try to be in band of stability after adjustment of their $\frac{n}{p}$ ratio



Case-I when $\frac{n}{p}$ ratio of isotope is greater than that required for stability.

→ stability will be gained by losing β-particle.

$$_0n^1 \longrightarrow {}_1H^1 + {}_{-1}e^0$$

(β-particle)



→ stability will be gained by losing

β-particle.

$$_0n^1 \longrightarrow {}_1H^1 + {}_{-1}e^0$$

(β-particle)

e.g $_6C^{14} \longrightarrow {}_7N^{14} + {}_{-1}e^0$

(β-emission)



$${}_0n^1 \longrightarrow {}_1H^1 + {}_{-1}e^0$$

Proton (β-particle)

e.g

$${}_6C^{14} \longrightarrow {}_7N^{14} + {}_{-1}e^0 \quad (\beta\text{-emission})$$

$${}_{15}P^{32} \longrightarrow {}_{16}S^{32} + {}_{-1}e^0 \quad (\beta^-)$$

Parent element

daughter element (stable)

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#2-MAGIC NUMBER| BAND OF STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

Symbols of nuclear particles,

$_0n^1$ → neutron

$_1H^1$ → Proton

$_1H^2$ → deuteron

$_1H^3$ → Triton

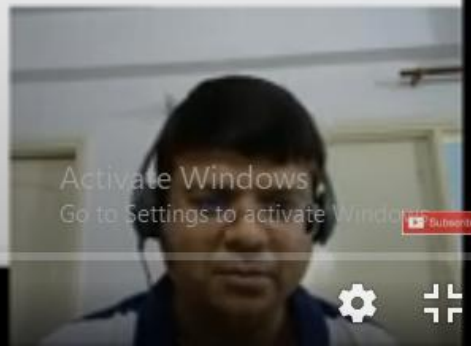
$_2He^4$ ⇝ α-particle

$_{+1}e^0$ ⇝ Positron

$_{-1}e^0$ ⇝ β-particle.

* Positron and β-particle are antiparticles.

*:



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18:33 / 30:00

$_1^2H$ → deuteron

$_1^3H$ → Triton

$_{-1}^{0}e$ → β-particle

* Positon and β-particle are antiparticles.
* β-particle is a nuclear particle whose mass & charge are exactly similar to that of electron.



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Similar to that of electron.

Case-II when $\frac{n}{p}$ ratio of isotope is smaller than that required for stability.

→ $\frac{n}{p}$ will be adjusted by any of the



#2-MAGIC NUMBER| BAND OF STABILITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYM...

→ $\frac{n}{p}$ will be adjusted by any of the following.

(a) By α-particle emission (observed in natural radioisotopes) (or heavier isotopes)

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21:19 / 30:00

(a) By α-particle emission (observed in natural radioisotopes) (or heavier isotopes)

e.g. $_{92}U^{235} \longrightarrow {}_{90}Th^{231} + {}_{2}He^{4}$

(α particle)



(or heavier isotopes)

e.g. $_{92}U^{235} \longrightarrow {}_{90}Th^{231} + {}_{2}He^{4}$

(α particle)

isodiaphers (i.e. A-2Z are same)

$235 - 2\times92 = 235 - 184 = 51$ | $231 - 2\times90 = 51$

emission of alpha particle always makes the daughter and parent elements isodiaphers



$235 - 2\times92$
$= 235 - 184$
$= \underline{51}$

$231 - 2\times90$
$= \underline{51}$

(b) By emission of positron : (observed in artifical radioisotopes)
↳ specially in lighter isotopes)

e.g. ${}_{7}^{13}N \longrightarrow {}_{6}^{13}C + {}_{+1}^{0}e$ (β - emission)

$\frac{n}{p} = \frac{6}{7}$ $\quad \frac{n}{p} = \frac{7}{6}$

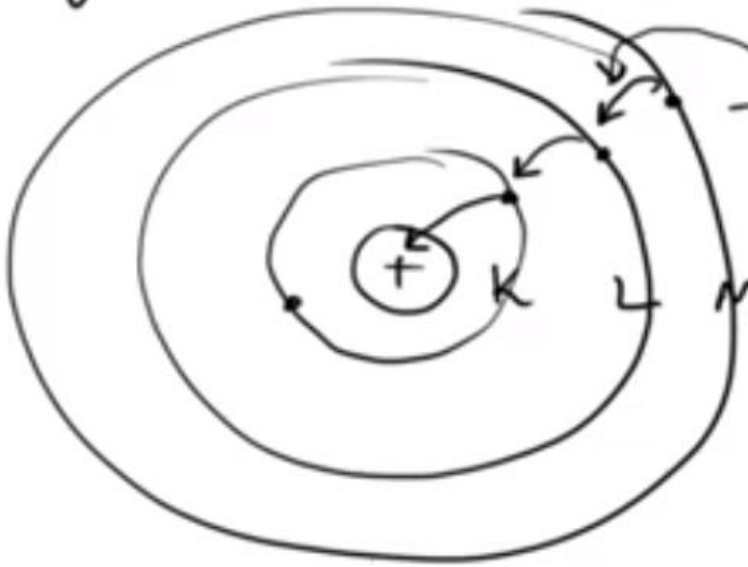


$\frac{n}{p} = \frac{6}{7}$ $\frac{n}{p} = \frac{7}{6}$ (β^- - emission)

(c) By K- electron capture : In some radioisotopes stability is gained by K-shell e^- capture by nucleus.

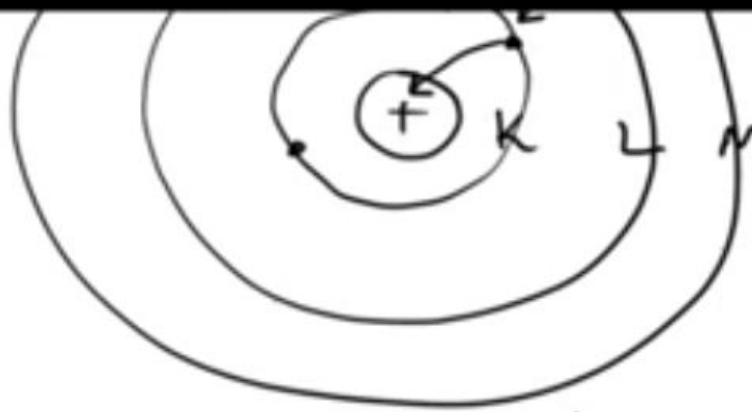
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stability is gained by K-shell e^- capture by nucleus.



- from these electronic transitions x-rays are emitted.


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transitions x-rays are emitted.

e.g. $_{19}K^{40} + {}_{-1}e^{0} \longrightarrow {}_{18}Ar^{40}$

($_{-1}e^{0}$: (K-shell e^- is captured); $_{18}Ar^{40}$: daughter element)

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(K-shell e^- is captured) daughter element

Q → which particle may be emitted to gain stability by $_{13}Al^{29}$.

Sol^n.

(A) β-particle (B) positron (C) α-particle (D) neutron.

Stable Al is $_{13}Al^{27}$, $\frac{n}{p} = \frac{14}{13}$

For given $_{13}Al^{29}$, $\frac{n}{p} = \frac{16}{13}$

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stability by $_{13}Al^{29}$.

Sol^{n}. (A) β-particle (B) positron (C) α-particle (D) neutron.

Stable Al is $_{13}Al^{27}$, $\frac{n}{p} = \frac{14}{13}$

For given $_{13}Al^{29}$, $\frac{n}{p} = \frac{16}{13}$

∴ β-particle emission Ans.



#3-THEORY OF RADIOACTIVE DISINTEGRATION|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12...

Theory of radioactive disintegration :

(Given by Rutherford and Soddy)



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0:43 / 12:37

(Given by Rutherford and Soddy)

According to this theory

(1) Nuclear disintegration is spontaneous and rate of nuclear disintegration does not depend on temp, physical state or free/combided form.

It is constant always.

$U^{235}_{(s)}$ / UF_6^{235} (g)

rate of nuclear disintegration is always constant



#3-THEORY OF RADIOACTIVE DISINTEGRATION|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12...

It is constant always. $U^{235}_{(s)} / UF^{235}_{6(g)}$

(i) During nuclear disintegration, parent element converts into a new element (called daughter element) having different physical and chemical properties than the parent element.



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4:27 / 12:37

(called daughter element) having different physical and chemical properties than the parent element.

e.g. $_{6}C^{14} \longrightarrow {}_{7}N^{14} + {}_{-1}e^{0}$ (β-decay)



#3-THEORY OF RADIOACTIVE DISINTEGRATION|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12...

(called daughter element) having different physical and chemical properties than the parent element.

e.g. $_{6}C^{14} \longrightarrow {}_{7}N^{14} + {}_{-1}e^{0}$ (β-decay)

Parent element — daughter element



5:28 / 12:37

Parent element daughter element

e.g $_{z}X^{A} \longrightarrow {}_{z-2}Y^{A-4} + {}_{2}He^{4}$

(α- particle)

In general, daughter element is 2 groups behind than the parent element.



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parent element in P.T.

(β - decay) $\quad {}_{Z}P^{A} \longrightarrow {}_{Z+1}Q^{A} + {}_{-1}e^{0}$

(β-particle)

daughter element will be shifted one group right in P.T. (in general)


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right in P.T. (in general)

Q→ Find no. of α and β-particles decayed in this conversion?

$$_{92}U^{238} \longrightarrow {}_{82}Pb^{206}$$

Sol^{n}: First calculate no of α-particles and β-particles.

α-particles : 8

β-particles = 6

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#4-RADIOACTIVE DISINTEGRATION SERIES|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLY...

Radioactive disintegration series :

→ All the isotopes having $Z > 83$ (i.e. beyond bismuth ($Z = 83$)) are found to be radioactive

→ Radioactive disintegration series consists of the starting elemnt to the end element (non-radioactive) including all the



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(1) Thorium Series : ($4n$ series)

called $4n$ because when mass no. of any element of the series is divided by 4, no remainder is left.

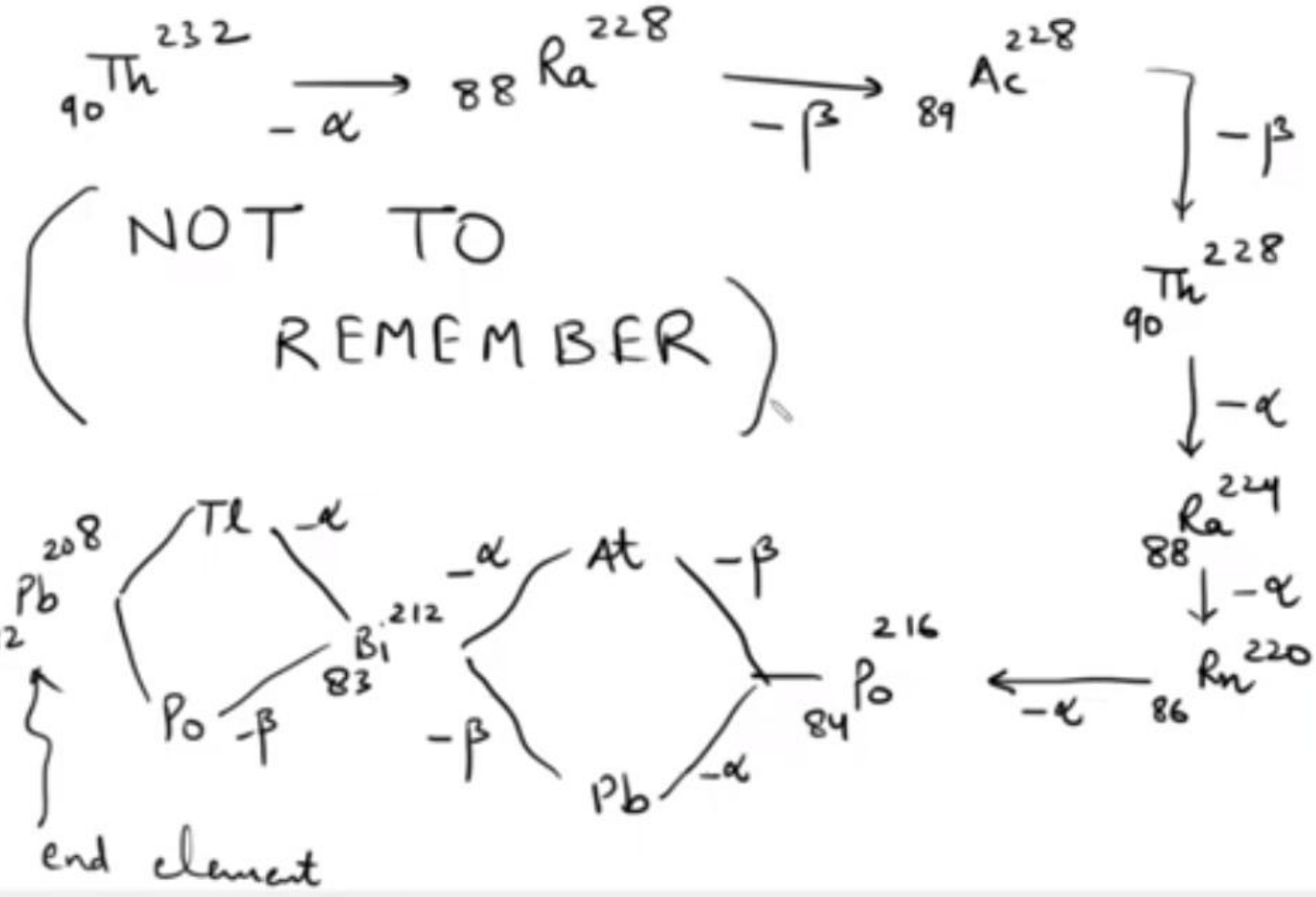
Starting element : $_{90}Th^{232}$

End element (non-radioactive) : $_{82}Pb^{208}$

} 6 α and 4 β particles emitted out.

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(NOT TO REMEMBER)



Called 4n because when mass no. of any element of the series is divided by 4, no remainder is left.

(i) Thorium Series : ($4n$ series)

(Natural Series)

Starting element : $_{90}Th^{232}$

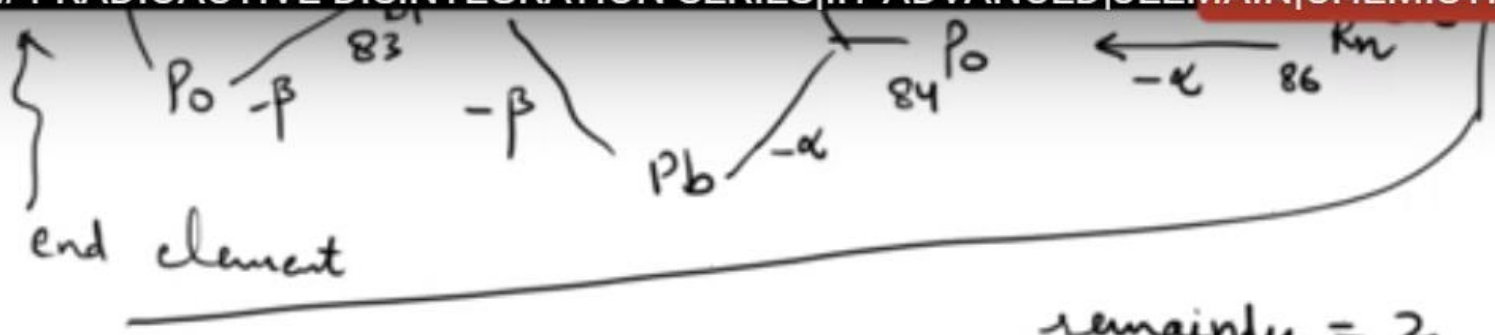
End element (non-radioactive) : $_{82}Pb^{208}$

} 6 α and 4 β particles emitted out.

$$_{90}Th^{232} \xrightarrow{-\alpha} {}_{88}Ra^{228} \xrightarrow{-\beta} {}_{89}Ac^{228} \xrightarrow{-\beta} \ldots$$

NOT TO

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remainder = 2

(ii) Uranium Series : (4n+2)

(Natural series)

$_{92}U^{238} \xrightarrow[6\beta \text{ emitted}]{8\alpha \text{ and}} {}_{82}Pb^{206}$

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$$_{92}U^{238} \xrightarrow[6\beta \text{ emitted}]{8\alpha \text{ and}} {}_{82}Pb^{206}$$

(iii) Actinium Series $(4n+3)$

(Natural

$$_{92}U^{235} \xrightarrow[4\beta \text{ particls}]{7\alpha \text{ and}} {}_{82}Pb^{207}$$

earlier it was thought to be an isotope of actinium.

Natural Series: starting element is naturally available

Hence, called actinium series



^{92} ↑ 4β particles ^{82}

earlier it was thought to be an isotope of actinium.

(iv) Neptunium Series $(4n+1)$

(Artificial Series)

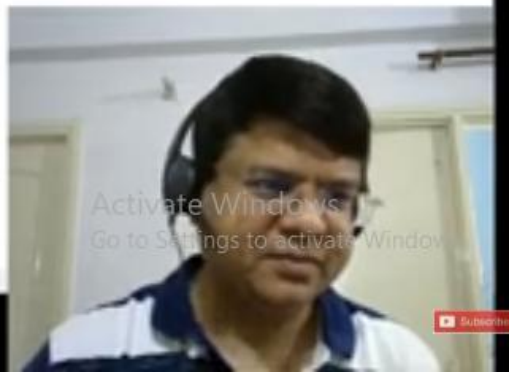
$^{237}_{93}Np \xrightarrow[4\beta \text{ particles emitted}]{7\alpha \text{ and}} {}^{209}_{83}Bi$

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emitted.

Comparison of α, β and γ - rays :

Property	α - rays	β - rays	γ - rays
nature	${}_{2}He^{4}$ (α-particles)	${}_{-1}e^{0}$ (β-particle)	EM radiation
velocity	relatively low $(1.4 \text{ to } 1.7) \times 10^{7}$ m/sec (depending upon source)	greater than α-particles	3×10^{8} m/sec (light)



#4-RADIOACTIVE DISINTEGRATION SERIES|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLY...

nature	$_2^4He$ (α-particles)	$_{-1}^0e$ (β-particle)	EM radiation
velocity	relatively low $(1.4 \text{ to } 1.7) \times 10^7$ m/sec (depending upon source)	greater than α-particles	3×10^8 m/sec (light)
Ionising Power	very high	lesser than α-particles	very low
Penetration power. (Size factor)	low	intermediate (much greater than α-)	Ver...

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Artificial Nuclear Reaction :

↳ New possibilities emerge when nuclei of some isotopes are bombarded with nuclear particles. Most important artificial nuclear rex is neutron induced fission.



NUCLEAR CHEMISTRY BY ALK SIR

Some isotopes are bombarded with nuclear particles. Most important artificial nuclear rex is neutron induced fission.

Representation of artificial nuclear rex :

e.g $_{4}Be^{9} + {}_{2}He^{4} \longrightarrow {}_{6}C^{12} + {}_{0}n^{1}$

(α-particle) (neutron)



Representation of artificial nuclear rex :

e.g $_{4}Be^{9} + {}_{2}He^{4} \longrightarrow {}_{6}C^{12} + {}_{0}n^{1}$

(α-particle) (neutron)

(Proj

OR

$_{4}Be^{9}(\alpha, n)\,{}_{6}C^{12}$

α : Projectile; n : emitted particle

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Projectile

emitted particle

Types of Artificial Nuclear Reaction :

(a) **Fission** (or Nuclear fission) : Most important nuclear fission is neutron induced (projectile) fission.



(a) **Fission** (or Nuclear fission) : Most important nuclear fission is neutron induced (projectile) fission.

C.f. $_{92}U^{235} + {}_0n^1 \longrightarrow {}_{56}Ba^{144} + {}_{36}Kr^{89} + 3\,{}_0n^1$

$_0n^1 \rightarrow U^{235}$

U^{235}

(chain reaction)

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C.J. $_{92}U^{235} + {}_{0}n^{1} \longrightarrow {}_{56}Ba^{144} + {}_{36}Kr^{89} + 3\,{}_{0}n^{1}$

$3\,{}_{0}n^{1} \rightarrow U^{235}$

$3\,{}_{0}n^{1} \rightarrow U^{235}$

(chain reaction)

→ Controlled nuclear fission is used as nuclear power plant

→ Uncontrolled nuclear fission is atom bomb (neutron induced)



(b) **Nuclear fusion** : Lightu nuclei (of H, He etc) do combine n fuse to form larger nuclei and release a huge amount of energy.

→ Energy generation in stars (e.g sun) is due to nuclear fusion.



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→ Energy generation in stars (e.g sun) is due to nuclear fusion.

on sun

$$ {}_1H^1 + {}_1H^1 \longrightarrow {}_1H^2 + {}_{+1}e^0 + \text{Energy} $$

(Position)

$$ {}_1H^2 + {}_1H^2 \longrightarrow {}_2He^3 + {}_0n^1 + \text{Energy} $$

$$ {}_1H^2 + {}_1H^2 \longrightarrow {}_1H^2 + {}_1H^1 + \text{Energy} $$

and so on ---

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and so on. - - -

→ Hydrogen bomb is based on the concept of nuclear fusion.

the temp. required is so much that to initiate a hydrogen bomb, an atom bomb must be set off first

↳ To ~~need~~ initiate a nuclear fusion a high temp. is needed to overcome the coulombic repulsion barrier.



fusion a high temp.
to overcome the coulombic repulsion barrier.

Q → Assuming that the charged particles emitted during natural decay of $_{92}U^{235}$ atoms are α and β- particles only. which of the following product(s) is/are not possible.

(A) $_{89}Ac^{231}$ (B) $_{89}Ac^{227}$ (C) $_{89}Ac^{225}$ (D) $_{82}Pb^{207}$



Q → Atomic No. of a radioactive element is 100. It first decays into an element Y and then decays into Z. In both the processes a charged particle is emitted. Which can be true?

(A) Y : Atomic No : 102

(B) Y : at. No : 101

(C) Z : At No : 100

(D) Z : At No : 99

Soln : (B) one β^-

(C) one β^- & then positron

(D) First α and β^-



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Q→ The no. of neutrons emitted when $_{92}U^{235}$ undergoes controlled nuclear fission to $_{54}Xe^{142}$ and $_{38}Sr^{90}$ is:

Sol^n:

$$_{92}U^{235} + {}_0n^1 \longrightarrow {}_{54}Xe^{142} + {}_{38}Sr^{90} + 4\,{}_0n^1$$

Ans: 4 Ans

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Rate of radioactive decay :

Experimentally it is observed that rate of nuclear decay (or radioactive) decay is directly proportional to number of nuclides of radioactive element at any instant.

(Follows First order kinetics



Rate of disintegration $\propto N_t$

$\Updownarrow$

or rate of decay or

radioactivity or activity

$$-\frac{dN_t}{dt} \propto N_t$$



or rate of decay or

radioactivity or activity

$$-\frac{dN_t}{dt} \propto N_t$$

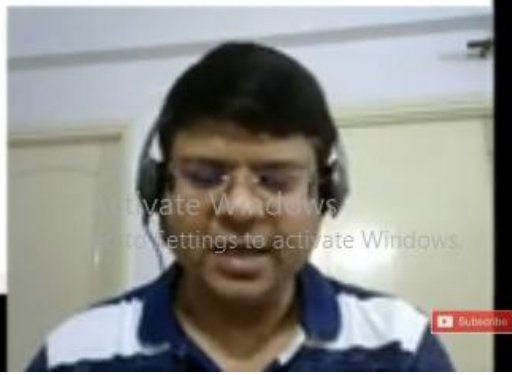
no. of nuclei or nuclides of radioactive substance

$$-\frac{dN_t}{dt} = \lambda \times N_t = \lambda \times \left(\frac{w}{M} \times N_A\right)$$

Ip

decay constant

(Unit : $time^{-1}$)



#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

$$-\frac{dN_t}{dt} = \lambda \times N_t = \lambda \times \left(\frac{w}{M} \times N_A\right)$$

Ip

decay constant

(Unit : $time^{-1}$)

$$N_t = N_0 e^{-\lambda t}$$

where N_0 = No of ~~nuclei~~ nuclides at $t = 0$.

N_t = No. of nuclides remaining at time = t.

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6:13 / 24:42

#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

N_t = No. of nuclides at time = t.

$$\lambda = \frac{1}{t} \ln\left(\frac{N_0}{N_t}\right)$$

Half life, $\boxed{t_{1/2} = \frac{\ln 2}{\lambda}} = \frac{0.693}{\lambda}$

Note :

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6:56 / 24:42

#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

Half life, $t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$

Note: Arrhenius equation does not apply on radioactive decay as radioactive decay does not depend on temperature. (spontaneous decay)



8:28 / 24:42

#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

radioactive decay as radioactive decay does not depend on temperature.
(spontaneous decay)

$$\text{Average life } (T_{av}) = \frac{\text{Total life of all nuclides}}{\text{Total no. of nuclides}}$$

$$= \int_0^{\infty} \frac{(-dN_t \times t)}{N_0}$$

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9:34 / 24:42

Average life (T_{av}) = $\frac{\text{Total life of all nuclides}}{\text{Total no. of nuclides}}$

$$= \int_0^{\infty} \frac{(-dN_t \times t)}{N_0}$$

$$T_{av} = \frac{1}{\lambda} \Rightarrow t_{y_2} = \frac{\ln 2}{\lambda} \Rightarrow$$

$$t_{y_2} = 0.693 \times T_{av}$$

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$$N_t = N_0 e^{-\lambda \times t}$$

if $t = T_{av}$, $\quad N_t = N_0 e^{-\lambda \times \frac{1}{\lambda}}$

$$= \frac{N_0}{e}$$

% of nuclides remain after T_{av}

$$= \frac{N_t}{N_0} \times 100 = \frac{1}{e} \times 100$$

$$= 36.79\ \%$$



#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

If $t = T_{av}$, $N_t = N_0 e^{-1}$

$$= \frac{N_0}{e}$$

% of nuclides remain after T_{av}

$$= \frac{N_t}{N_0} \times 100 = \frac{1}{e} \times 100$$

$$= 36.79\ \%$$

% of nuclides decayed in T_{av} time

$$= (100 - 36.79)$$

$$= 63.21\ \%$$

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12:34 / 24:42

$= \frac{}{N_0} \times 100$ e

$= 36.79\%$

% of nuclides decayed in T_{av} time

$= (100 - 36.79)$

$= 63.21\%$

Note: ② Specific activity is the activity or radioactivity per unit mass (g) of radioactive substance.



#6-RATE OF RADIOACTIVE DECAY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPIAD|K...

$= \frac{}{N_0} \times$ e

$= 36.79\%$

% of nuclides decayed in T_{av} time

$= (100 - 36.79)$

$= 63.21\%$

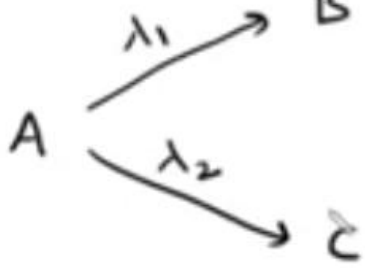
Note: (2) Specific activity is the activity or radioactivity per unit mass (g) of radioactive substance.

↳ Activity is measured by Zeiger-Muller c
in dps or dpm (disintegration per minute)



14:49 / 24:42

Parallel decay :



$$\lambda = \lambda_1 + \lambda_2$$

$$\frac{0.693}{t_{y_2}} = \frac{0.693}{(t_{y_2})_1} + \frac{0.693}{(t_{y_2})_2}$$

$$\left\{ \frac{[B]}{[C]} \right\}_t = \frac{\lambda_1}{\lambda_2}$$



→ No. of nuclides remaining after

x half lives $= \frac{N_0}{2^x}$.

→ Sequential or consecutive decay:

$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C \text{ - - - - - } P$$

A: Parent element

B: intermediate

P: Non-radi... element



$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C - - - - - P$$

A: Parent element

B: intermediate

P: Non-radioactive element

→ time at which no. of nuclides of intermediate 'B' is given by,

$$t_{max} = \frac{\ln(\lambda_2/\lambda_1)}{(\lambda_2 - \lambda_1)}$$



$$A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C \longrightarrow \cdots \longrightarrow P$$

Rate of formation of B = Rate of decay of B

(for an intermediate)

⇕

Radioactive equilibrium.

$$\lambda_A N_A = \lambda_B N_B .$$

$$\frac{N_A}{N_B} = \frac{\lambda_B}{\lambda_A} = \frac{\ln 2 / (t_{y_2})_B}{\ln 2 / (t_{y_2})_A}$$

$$\Rightarrow \boxed{\frac{N_A}{N_B} = \frac{(t_{y_2})_A}{(t_{y_2})_B}}$$



#7-UNITS OF RADIOACTIVITY|SPECIFIC ACTIVITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS ...

Units of radioactivity or activity :

SI unit : Bacqueral (or Bq)

1 Bq = 1 dps (or 1 cps)

dps ↑ disintegration per second

cps ↑ count per second.

⟶ Activity of a radioactive substance is measured in Geiger-Muller counter.



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1:31 / 18:26

NUCLEAR CHEMISTRY BY ALK SIR

Imp

→ 1 curie (or 1 Ci) = 3.7×10^{10} dps

* 1 curie is the amount of radioactive substance which shows radioactivity equal to 3.7×10^{10} dps.

* 1 curie is measured by taking reference of Ra^{225}. 1 Ci is the activity of 1g of Ra^{225}.

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NUCLEAR CHEMISTRY BY ALK SIR

* 1 Curie is measured by taking reference of Ra^{225}. 1 Ci is the activity of 1g of Ra^{225}. $\left[(t_{y_2})_{Ra^{225}} = 1600 \text{ years}\right]$

Activity of Ra^{225} (1g) $= \lambda \times N$

$$= \frac{\ln 2 \times \left(\frac{1}{225} \times 6.02 \times 10^{23}\right)}{1600 \times 365 \times 24 \times 3600}$$

$$\cong 3.7 \times 10^{10} \text{ dps.}$$



NUCLEAR CHEMISTRY BY ALK SIR

$\cong 3.7 \times 10^{10}$ dps.

$\longrightarrow$ 1 Rutherford (or 1 Rd) $= 10^{6}$ dps.

* Specific activity : activity of 1 g of substance
or activity per unit mass.

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or activity per unit mass.

Ip

Q → The activity per ml of a solution of radioactive substance is x. How much water to be added to 200 ml of this solution so that activity falls to $\frac{x}{20}$ per ml after 4 half lives?

Sol^{n}: V ml water to be added.

$$\left(\frac{200x}{2^4}\right) = \frac{x}{20}\underbrace{(200+V)}_{\text{total volume}}$$



after 4 half lives?

Sol^{n}: V ml water to be added.

$$\left(\frac{200x}{2^4}\right) = \frac{x}{20}(200+V)$$

total volume after mixing water

$$\frac{200}{\frac{x}{4}} = \frac{1}{\frac{20}{5}}(200+V)$$

$V = 50$ ml Ans

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NUCLEAR CHEMISTRY BY ALK SIR

Q → A 0.1ml sample of a solution containing 10^{-7} Ci of $_1H^3$ (radioactive) is injected into the blood stream of a laboratory animal. within no time, circulatory equilibrium is established and 0.1ml of blood is found to have an activity of 20 dpm. Find blood volume of animal.

Sol^n : $10^{-7} \times 3.7 \times 10^{10}$ dps | 0.1ml blood → 20 dpm.



NUCLEAR CHEMISTRY BY ALK SIR

$$10^{-7} \times 3.7 \times 10^{10} = \frac{\cancel{20}}{\cancel{60}\ 3} \times \frac{1}{0.1} \times V$$

$$V = 0.3 \times 3.7 \times 1000$$

$$= 3 \times 370 \text{ ml}$$

$$= 1110 \text{ ml}$$

$$V_{\text{blood of animal}} = 1110 - 0.1$$

$$= \underline{1109.9 \text{ ml}}$$



NUCLEAR CHEMISTRY BY ALK SIR

Applications of Radioactivity :

1. Rock dating (or Helium dating) :
2. Carbon dating
3. In medicine (in cancer therapy) :

↳ e.g PET, F^{18}, C^{11}, N^{13} type of radioactive substance is injected in blood stream to detect tum



NUCLEAR CHEMISTRY BY ALK SIR

↳ e.g PET, F, C, N type of radioactive substance is injected in blood stream to detect tumour.

(3) To study dynamic equilibrium nature

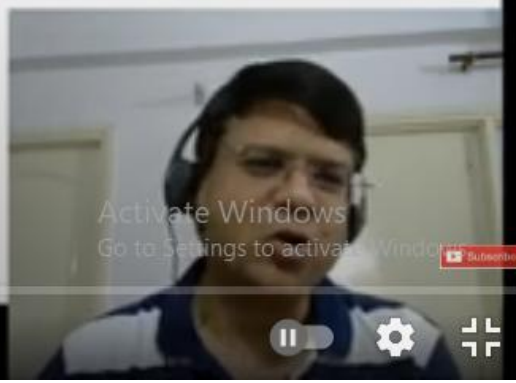


#8-APPLICATIONS OF RADIOACTIVITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPI...

① Rock dating (or Helium dating)

⟶ To determine age of rocks/minerals/ores.
(non-living)

e.g Th ores / U ores etc.



5:27 / 28:21

(non-living)

e.g Th ores / U ores etc.

e.g a Thorium ore,

$$_{90}Th^{232} \xrightarrow{K} {}_{82}Pb^{208} + 6\,{}_{2}He^{4} + 4\,{}_{-1}e^{0}$$

(α-decay) (β)



NUCLEAR CHEMISTRY BY ALK SIR

(non-living)

e.g Th ores / U ores etc.

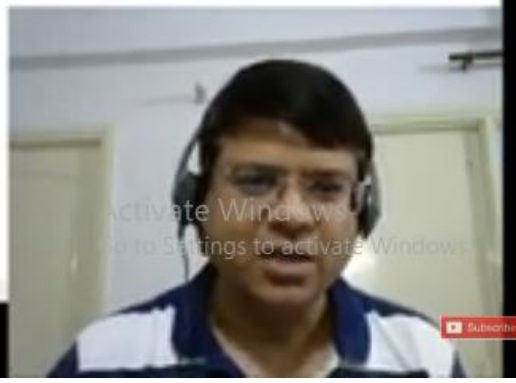
e.g a Thorium ore,

α-particles are assumed to be trapped in the rock as He gas.

$$^{232}_{90}Th \xrightarrow[\lambda]{x} {}^{208}_{82}Pb + 6\,{}^{4}_{2}He + 4\,{}^{0}_{-1}e$$

(α-day) (β)

$t=0$,	a	0	0	0
$t=t_{age}$,	$a-x$	x	$6x$	



NUCLEAR CHEMISTRY BY ALK SIR

e.g a Thorium ore, → to be trapped in the rock as He gas.

$$^{232}_{90}Th \xrightarrow[\lambda]{\alpha} {}^{208}_{82}Pb + 6\,{}^{4}_{2}He + 4\,{}^{0}_{-1}e$$

(α - day) (β)

$t=0$,	a	0	0	0
$t=t_{age}$,	$a-x$	x	$6x$	

$$t_{age} = \frac{1}{\lambda} \ln\left(\frac{a}{a-x}\right)$$



Q→ In nature a decay chain series starts with $_{90}Th^{232}$ and finally terminates at $_{82}Pb^{208}$. A Th ore sample was found to contain 8×10^{-5} ml of He at STP and 5×10^{-7} g of Th^{232}. Find age of ore sample and assume all helium to be due to decay of Th^{232}. $(t_{1/2})_{Th^{232}} = 1.4\times10^{10}$ years.

Soln:

$$_{90}Th^{232} \longrightarrow {}_{82}Pb^{208} + 6\ {}_{2}He^{4} + 4\ {}_{-1}e^{0}$$

$t=0$	a	0	0
$t=t_{age}$	$a-x$	x	$6x$

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NUCLEAR CHEMISTRY BY ALK SIR

be due to decay of Th. $(t_{1/2})_{Th}$

Sol^n :

$$^{232}_{90}Th \longrightarrow {}^{208}_{82}Pb + 6\,{}^{4}_{2}He + 4\,{}^{0}_{-1}e$$

$t=0$	a	0	0
$t=t_{age}$	$a-x$	x	$6x$

$6x = \frac{8\times10^{-5}}{22400}$

$x =$ ✓

$(a-x) = \frac{5\times10^{-7}}{232}$ mol

$$t_{age} = \frac{1}{\frac{\ln 2}{(1.4\times10^{10})}} \times \ln\frac{\left(\frac{5\times10^{-7}}{232} + \frac{8\times10^{-5}}{6\times22400}\right)}{\left(\frac{5\times10^{-7}}{232}\right)}$$

$$= 4.9\times10^{9} \text{ years}$$



#8-APPLICATIONS OF RADIOACTIVITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPI...

$\left(\frac{1.4\times10^{10}}{232}\right)$

$= 4.9 \times 10^{9}$ years Ans.

He dating for inorganic materials (rocks and ores) (non-living)

Carbon dating is used for organic materials (which were once living)

(2) Carbon dating: (C^{14} is used to determine age of organic materials like animal fossils or wooden logs)



17:25 / 28:21

age of organic materials like animal fossils or wooden logs)

→ C^{14} is produced in upper atmosphere as

$$_{7}N^{14} + {}_{0}n^{1} \longrightarrow {}_{6}C^{14} + {}_{1}H^{1}$$

($_{0}n^{1}$: neutron captured by nitrogen)

($_{1}H^{1}$: Protons)

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→ C^{14} is produced in upper atmosphere as

$$_{7}N^{14} + {}_{0}n^{1} \longrightarrow {}_{6}C^{14} + {}_{1}H^{1}$$

($_{0}n^{1}$: neutron captured by nitrogen)

($_{1}H^{1}$: Protons)

$_{6}C^{14}$: Radioactive (undergoes β- decay with half life nearly 5770 years)



#8-APPLICATIONS OF RADIOACTIVITY|IIT ADVANCED|JEEMAIN|CHEMISTRY CLASS 12|OLYMPI...

→ C^{14} diffuses into our atmospheres and gets absorbed into plants through photosynthesis (to a certain concentration). Also C^{14} is retained in living animal in a certain concentration as radioactive eq^{n} is established.

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21:55 / 28:21

concentration as radioactive eq^{m} is established.

→ Conc. of C^{14} remains constant in living plant and living animals.

Rate of absorption of C^{14} = Rate of decay of C^{14}

(Radioactive equilibrium)



NUCLEAR CHEMISTRY BY ALK SIR

$${}_{6}C^{14} \longrightarrow {}_{7}N^{14} + {}_{-1}e^{0}$$ (β- decay), $(t_{y_2})_{C^{14}} = 5770$ years.

To determine age of fossil or wooden log,

$$t_{age} = \frac{1}{\frac{\ln 2}{(t_{y_2})_{C^{14}}}} \times \ln \frac{(C_{C^{14}})_0}{(C_{14})_t}$$

